

Introduction to Auctions

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What is an action?

- An auction is a mechanism used by a seller to allocate an object to N potential buyers.
- The seller does not know each buyer's willingness to pay.
- Thus, instead of the sellers making extending an offer to the buyers or engaging in negotiations, the buyers make competing bids.
- Since bidders do not know each other's valuations, it is a game on incomplete information.
- The rules of the action determine the winner as well as the sale price.

Common Auctions

- **Second Price Auction (SPA):** bidders simultaneously submit bids. The highest bid wins the object and pays the *second highest bid*.
- **English Action (EA):** Ascending bid auction in which bid keeps increasing until only one bidder remains. He pays the price at which the last competitor dropped.
- **First Price Auction (FPA):** Bidders simultaneously submit sealed bids. The highest bid wins the object and pays the *his bid*.
- **Dutch Action (DA):** The price declines continuously until one of the bidders stops the auction and pays the price at which the auctions was stopped.

Assumptions

- Each bidder i privately knows his own value.
- Valuations are independent draws from a common distribution with cdf $F(\cdot)$.
- Each bidder is risk neutral expected utility maximizer.
- The seller is a risk neutral expected utility maximizer and possesses valuation v_S for the object, which is common knowledge.
- Ties between winning bids are resolved randomly.
- The seller has a choice of setting
 - entry fee $c \geq 0$ (cost of participating in the auction).
 - reserve price $r \geq 0$ (minimum bid).

Second Price Auction $SPA(r, c)$

- Set of strategies available to bidder i : $S_i = \{No\} \cup [r, \infty)$.
- A strategy for i is a bid $b_i : v_i \rightarrow S_i$.
- Let $\tilde{y}_i = \max\{v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_N\}$.
- Let $\tilde{b}_i = \max\{b_1, \dots, b_{i-1}, b_{i+1}, \dots, b_N\}$
- Bidder i ex-post payoff:

$$u_i(b_i, v_i, \tilde{b}_i) = \begin{cases} 0 & \text{if } b_i < \tilde{b}_i \\ (v_i - \tilde{b}_i)p(b_i, \tilde{b}_i) & \text{if } b_i = \tilde{b}_i \\ v_i - \tilde{b}_i & \text{if } b_i > \tilde{b}_i \end{cases}$$

where $p(b_i, \tilde{b}_i)$ is the probability of i winning the object in case of a tie.

Second Price Auction with No Entry Fee: $SPA(r, 0)$

Proposition 1

In the $SPA(r, 0)$, each bidder i has a weakly dominant strategy:

$$b_i^H(v_i) = \begin{cases} \text{No} & \text{if } v_i < r \\ v_i & \text{if } v_i \geq r \end{cases} \quad (1)$$

Proof

- $v_i < r$ - bidding $b_i \geq r$ results in at most 0 payoff and thus not bidding is at least as good.

Second Price Auction with No Entry Fee: $SPA(r, 0)$

Proof

- $v_i \geq r$: two other possibilities:
 - $b_i > v_i$ results in payoff

$$(v_i - \tilde{b}_i)Pr(b_i > \tilde{b}_i)$$

From ex-ante point of view, with probability $Pr(\tilde{b}_i \leq v_i)$ the buyer's payoff is positive and with prob. $Pr(\tilde{b}_i > v_i | b_i > \tilde{b}_i)$ is negative. It is straightforward to verify that $b_i = v_i$ performs at least as good since it result in the same payoff with prob. $Pr(\tilde{b}_i \leq v_i)$ and zero payoff otherwise.

- By the same token, it is straightforward to verify that $b_i = v_i$ dominates $b_i < v_i$ for any strategy of the other bidders.

Second Price Auction with Entry Fee: $SPA(r, c)$

- Upon entry, it is still a weakly dominant strategy to bid $b_i = v_i$.
- However, for $c > 0$ not all bidders would choose to enter.
 - $v_i = r$ - optimal not to enter.
- Suppose that bidders follow a cut-off strategy: No if $v_i < v_0$ and $b_i(v_i) = v_i$ if $v_i \geq v_0$.
- Bidder's i expected payoff will be:

$$Pr(\tilde{y}_i < v_i)v_i - Pr(\tilde{y}_i \in [v_0, v_i])E[\tilde{y}_i | \tilde{y}_i \in [v_0, v_i]] - Pr(\tilde{y}_i < v_0)r - c$$

- You can easily show that this expected payoff is increasing in v_i and for $v_i = v_0$ it should satisfy.

$$Eu_0 = Pr(\tilde{y}_i < v_0)(v_0 - r) - c = 0 \quad (2)$$

- Let $G(\cdot)$ denote the distribution of $\tilde{y}$.
- Then $G(v_i) = Pr(\tilde{y}_i < v_i) = \prod_{j \neq i} Pr(v_j < v_i) = F(v_i)^{N-1}$.

Second Price Auction with Entry Fee: $SPA(r, c)$

- Then we can rewrite equation (2) as

$$G(v_0)(v_0 - r) = c \quad (3)$$

- Note that for $c = 0$, $v_0 = r$.
- In a symmetric BNE,

$$b_i^H(v_i) = \begin{cases} \text{No} & \text{if } v_i < v_0(r, c) \\ v_i & \text{if } v_i \geq v_0(r, c) \end{cases} \quad (4)$$

First Price Auction $FPA(0, 0)$

- Intuitively, bidding one's own valuation is not a BNE anymore since it results in 0 payoff.
- Bidder i ex-post payoff:

$$u_i(b_i, v_i, \tilde{b}_i) = \begin{cases} 0 & \text{if } b_i < \tilde{b}_i \\ (v_i - b_i)p(b_i, \tilde{b}_i) & \text{if } b_i = \tilde{b}_i \\ v_i - b_i & \text{if } b_i > \tilde{b}_i \end{cases}$$

- Bidder i strategy $b_i : v_i \rightarrow \{No\} \cup [0, \infty)$.

Deriving the Bidding Function for $FPA(0, 0)$

- Let us suppose that all $j \neq i$ follow a symmetric bidding function $b(v)$, which is also strictly increasing in v and differentiable.
- Moreover, it must be true that $b(0) = 0$.
- Bidder i optimization problem becomes:

$$\max_{b_i} Pr(b_i > \max\{b(v_1), \dots, b(v_{i-1}), b(v_{i+1}), \dots, b(v_N)\})(v_i - b_i)$$

- Note that $b(v)$ is invertible. Therefore, $b_i > b(v) \Leftrightarrow b^{-1}(b_i) > v$. Thus, i 's optimization problem can be written as

$$\max_{b_i} G(b^{-1}(b_i))(v_i - b_i)$$

$$FOC : g(b^{-1}(b_i(v_i))) \frac{db^{-1}(b_i)}{db_i} (v_i - b_i(v_i)) - G(b^{-1}(b_i)) = 0$$

Deriving the Bidding Function for $FPA(0, 0)$

- Note that $b(b^{-1}(b_i)) = b_i$. Differentiating both sides, we get:

$$b'(b^{-1}(b_i)) \frac{db^{-1}(b_i)}{db_i} = 1 \rightarrow \frac{db^{-1}(b_i)}{db_i} = \frac{1}{b'(b^{-1}(b_i))}$$

- Therefore,

$$FOC : \frac{g(b^{-1}(b_i))}{b'(b^{-1}(b_i))} (v_i - b_i) - G(b^{-1}(b_i)) = 0$$

- For a symmetric equilibrium, it must be true that $b_i(v_i) = b(v_i)$.
- Thus, imposing symmetry, we obtain $b^{-1}(b(v_i)) = v_i$ and

$$\frac{g(v_i)}{b'(v_i)} (v_i - b(v_i)) - G(v_i) = 0$$

Deriving the Bidding Function for $FPA(0, 0)$

- Solution of this first degree differential equation with $b(0) = 0$ is

$$b^I(v_i) = \frac{\int_0^{v_i} y_i g(y_i) dy_i}{G(v_i)} = E[\tilde{y}_i | \tilde{y}_i < v_i] \quad (5)$$

- Thus, the bidder always bids the expected value of the highest valuation of the other bidders conditional on his valuation being the highest.
- Integrating by parts the above equation:

$$b^I(v_i) = v_i - \int_0^{v_i} \frac{G(y_i)}{G(v_i)} dy_i$$

- Thus, the $FPA(0,0)$ always results in bid-shedding.
- Note that $\frac{G(y_i)}{G(v_i)} = \left[\frac{F(y_i)}{F(v_i)} \right]^{N-1}$
- Thus, $\lim_{N \rightarrow \infty} b_i(v_i) = v_i$.

Preferred Auction from Seller's Point of View

- SPA(0,0) leads to the bidder's placing bids equal to their valuations $b^H(v) = v$, but pay the second highest bid.
 - The equilibrium expected payment by a bidder is:

$$m^H(v) = \text{Prob}[\tilde{y} < v] \times E[\tilde{y} | \tilde{y} < v]$$

- FPA(0,0) leads to bid shedding with equilibrium bid of $b^I(v) = E[\tilde{y} | \tilde{y} < v]$.
 - Thus, the equilibrium expected payment by a bidder is:

$$m^I(v) = \text{Prob}[\tilde{y} < v] \times E[\tilde{y} | \tilde{y} < v] = m^H(v)$$

- The two auctions are revenue equivalent.

Auctioneer's revenue

- Let the support of $v \in [0, 1]$.
- The ex-ante expected payment of a bidder is:

$$E[m(v)] = \int_0^1 (\text{Prob}[\tilde{y} < v] \times E[\tilde{y} | \tilde{y} < v]) dv = \quad (6)$$

$$= \int_0^1 \left(\int_0^v yg(y) dy \right) f(v) dv \quad (7)$$

- Changing the order of integration:

$$\begin{aligned} E[m(v)] &= \int_0^1 \left(\int_y^1 f(v) dv \right) yg(y) dy = \\ &= \int_0^1 y(1 - F(y))g(y) dy \end{aligned}$$

Auctioneer's revenue

- Since bidders are symmetric, the expected revenue to the auctioneer is $E[R] = NE[m(v)] = \int_0^1 Ny(1 - F(y))g(y)dy$.
- This has nice interpretation using order statistics.
- Order Statistics:
 - If there are N draws from a distribution $F(\cdot)$, then $\{Y_1^{(N)}, \dots, Y_N^{(N)}\}$ is an arrangement of the draws so that $Y_1^{(N)} \geq Y_2^{(N)} \dots \geq Y_N^{(N)}$.
 - The random variables $Y_k^{(N)}$ is referred to order statistics.
 - First order statistics ($k = 1$) is simply the distribution of the maximum of all N draws

$$F_1^{(N)}(y) = Pr(Y_i \leq y)^N = F^N(y)$$

Order Statistics

- The second order statistics is the distribution of the second highest of all N draws.

$$\begin{aligned}F_2^{(N)}(y) &= Pr(\text{all } Y_i < y) + Pr(\text{all } Y_i \text{ except the highest are } < y) = \\&= F^N(y) + NF^{N-1}(y)(1 - F(y)) = \\&= NF^{N-1}(y) - (N - 1)F^N(y)\end{aligned}$$

- The pdf of the second order statistics is:

$$f_2^{(N)}(y) = N(N - 1)(1 - F(y))F^{N-2}(y)f(y)$$

- Recall that $G(y) = F^{N-1}(y)$ and $g(y) = (N - 1)F^{N-2}(y)f(y)$.

$$f_2^{(N)}(y) = N(1 - F(y))g(y)$$

- Therefore, $E[R] = \int_0^1 Ny(1 - F(y))g(y)dy = E[Y_2^{(N)}]$.

References

- V. Krishna "Auction Theory"- Ch. 2 and 3
- Matthews, Steven "A technical primer on auction theory I-Independent Private Values"- posted on eLearning